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### Fidelity of optimally controlled quantum gates with randomly coupled multiparticle environments

Matthew D. Grace<sup>a</sup>, Constantin Bif<sup>a</sup>, Herman Batte<sup>a</sup>, Daniel A. Lidar<sup>a</sup>, Jari A. Womslay<sup>b</sup> & Robert L. Head<sup>c,d</sup>

<sup>a</sup>Department of Chemistry, Princeton University, Princeton, New Jersey 08544, USA

<sup>b</sup>Departments of Chemistry, and Chemical Engineering and Physics, University of Southern California, Los Angeles, CA 90089, USA

<sup>c</sup>Department of Physics, University of Oxford, Oxford, OX1 3PU, UK

<sup>d</sup>5C, Solwin Drive, 2501 Oakwood Parkway, Sunnyvale, CA 94085, USA

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